

Learner Name

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1. Did you use any AI tool(s) to assist with this

☐ No. Why not?

☒ Yes. List the AI tool(s) you used and briefly describe the purpose for which you used the tool(s). (*e.g., brainstorming, research, drafting content*)

ChatGPT was used as a methodological reviewer, coding and debugging assistant, result interpreter, architecture-design assistant, and drafting aid. I used it to review the project protocol, generate and review Python code, interpret locally produced model/calibration/fairness outputs, design the Azure architecture, and help structure the report, Model Card, README, and AI disclosure. ChatGPT image generation was also used for the architecture diagram. All experiments and final evaluation were executed locally.

2. To demonstrate effective AI use, please provide your initial prompt(s) and a brief description of the steps you took to refine and iterate on the AI's output for this

3.

Initial prompt 1: “I’ve attached my final project and a brief I’d like to stick to in drafting this. Review both documents and flag any anomaly before we start drafting the final project.”

Refinement: The review identified that the Module 6 test set had already been inspected and should not be treated as an untouched final-project holdout. I revised the protocol before experimentation, created a new stratified 80/20 split with seed 20260920, restricted model selection/calibration/ablation to development data, and froze the pipeline before opening the 1,409-row holdout.

Model and calibration iteration: I shared locally generated terminal outputs and screenshots rather than reconstructing detailed prompts after the fact. AI helped interpret the results against pre-defined rules. Logistic regression, gradient boosting, and the ANN were practically tied on development PR-AUC, so I kept the simpler logistic model. The calibration hypothesis was not supported, so I reported the negative result instead of trying alternative methods post hoc.

Fairness iteration: After the protected-attribute ablation, AI helped separate the question of predictive value from the question of fairness. gender and SeniorCitizen were removed from predictive inputs because performance remained practically tied, but both were retained for group-level auditing.

Post-holdout refinement: After I supplied the final holdout output, AI noticed that the printed business summary used the exact threshold 0.398860 while the frozen operational rule was 0.40. I did not retrain or rerun the holdout. I recalculated the business result from the saved immutable probabilities at 0.40 and preserved the original evaluation artifact.

Initial prompt 2: “architecture diagram looks too noisy. I need a standard scientific paper type diagram that is academically acceptable. Not AI slop.” The first visual was rejected and replaced with a simpler academic systems diagram.

4. Reflect on the AI-generated content used in your submission:

What were its key strengths and weaknesses?

Strengths: AI accelerated protocol review, code drafting, debugging, interpretation, and documentation. It was especially useful as a second reviewer, including identifying the post-holdout threshold-reporting inconsistency and helping challenge unsupported assumptions about model complexity and calibration.

Weaknesses: its suggestions still required verification. Some initial expectations were not supported by the data, generated code contained a reporting inconsistency, and the first architecture image was visually unsuitable for an academic paper. AI could also make prose sound more certain than the evidence justified unless I explicitly constrained the wording.

How will you improve your prompting for similar tasks in the future to achieve better results?

For similar tasks I will provide the evaluation protocol, frozen decisions, artifact names, and reporting constraints at the start. I will ask AI to separate observed results, assumptions, and recommendations; state what evidence supports each claim; and flag any proposed change that would violate a locked protocol. For visuals, I will specify an academic technical-diagram style from the first prompt. I will also preserve exact prompts as I work instead of reconstructing the interaction later.

5. List at least one source you consulted to verify the AI-generated content in your work.

I verified AI-generated technical claims using primary or peer-reviewed sources and my own saved project artifacts. Examples include:

- IBM Cognos Analytics documentation for the Telco Customer Churn sample and dataset context.
- scikit-learn documentation for RepeatedStratifiedKFold, probability calibration, and Brier score.
- Saito & Rehmsmeier (2015), PLOS ONE, for precision-recall evaluation under class imbalance.
- Microsoft Learn documentation for Azure Machine Learning batch endpoints, MLflow model registration, Azure Data Factory, and Azure Databricks Feature Store.
- Local CSV/JSON artifacts and automated tests for all reported numerical results.

6. Did you directly incorporate AI output into your submission, without any modifications?

- ☐ No. Describe how you modified the AI-generated content and why those modifications were necessary.

- ☒ Yes. Reflect on your reasons for directly using the output. How could this change the nature of jobs that heavily require this type of work?

Yes, in limited cases. The revised AI-generated architecture diagram was used directly after I rejected the first version and reviewed the replacement against the approved technical design. Some AI-generated code was also retained when it passed local execution, automated tests, and output checks. Report prose was reviewed and adjusted for accuracy, evidence, scope, and consistency with the project artifacts.

I used these outputs directly only where the task was primarily implementation or presentation and where I could independently verify the result. The empirical results themselves were not generated by AI: training, cross-validation, holdout evaluation, fairness metrics, and business calculations were produced locally and preserved as project artifacts.

For jobs that rely heavily on coding, analysis, documentation, or technical communication, this type of AI can shift work away from producing every first draft manually and toward specifying requirements, reviewing outputs, validating evidence, detecting errors, and taking responsibility for final decisions. Domain knowledge and verification become more important, not less, because plausible AI output can still be incorrect or unsuitable.